ECON2200H

Introduction to Statistics for Economics and Management

2020 Winter Term

Week 04: Random Variables and Their Distributions

29 January 2020

Reading: Chapters 5 and 6 of textbook.

Outline

- Random variables
- Probability distribution
- Expected value, variance
- Some discrete probability distributions
- Some continuous probability distributions

1. Random variables

- Let S be the sample space of an experiment, a **random variable** X is a function from S to the real numbers.
- That is, a random variable assigns a numerical value to each possible outcome in the sample space.
- Because the outcome of an experiment is random, the real number produced by the function is also random.
- **Example**. Tossing a coin three times, sample space S = {HHH, HHT, HTT, HTH, TTT, TTH, THH, THT}.
    - The total number of heads is a random variable. It takes values 0, 1, 2, or 3.
- If outcome s is the sequence HTH, then $X(s) = 2$.
- If outcome s is the sequence TTT, then $X(s) = 0$.

- In the above example, the number of heads minus the number of tails is also a random variable. It takes values of -3, -1, 1, or 3.
- More examples of random variables:
    - The lifetime of a new battery.
    - The winning number of LOTT 6/49.
    - The amount of rainfall in Peterborough.

2. Discrete random variables and their distributions

- A random variable is **discrete** if it takes on only a finite number of values, or, at most, a countably infinite number of values.
- Random variables defined in the previous example are discrete.
- We are interested in the probability of a random variable.
- **Example**. Tossing a coin three times. Let X = the total number of heads. S = {HHH, HHT, HTH, HTT, TTT, TTH, THT, THH}.

Outcomes of event	{TTT}	{HTT, TTH, THT}	{HHT, HTH, THH}	{HHH}
x	0	1	2	3
$P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

- **Example** (continued). Let X = the number of heads minus the number of tails.

Outcomes of event	{TTT}	{HTT, TTH, THT}	{HHT, HTH, THH}	{HHH}
x	-3	-1	1	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

2-1. The probability mass function

- The **distribution** of a random variable X is the collection of probabilities $P(X = x)$ for all values of real variable x.
- The distribution of X shows all values of the pair $(x, P(X = x))$.
- For a discrete random variable X, the **probability mass function** of X is defined as the function f, such that for every real number x, $f(x) = P(X = x)$.
- Some also refer to the probability mass function as the **probability function**.

- The **support** of X is the set of real numbers x_i with $P(X = x_i) > 0$, for all i.
- Let $x_1, x_2, \cdots$ be the support of X, the probability mass function satisfies the following.

  1). $f(x) > 0$ if $x = x_j$ is in the support of X, and $f(x) = 0$ otherwise.

  2). $\sum_{i=1}^{\infty} f(x_i) = 1$.

- **Example**. Roll two dice. Let X be the sum of the two numbers that appear.

$$
\begin{array}{cccccc}
(1,1) & (2,1) & (3,1) & (4,1) & (5,1) & (6,1) \\
(1,2) & (2,2) & (3,2) & (4,2) & (5,2) & (6,2) \\
(1,3) & (2,3) & (3,3) & (4,3) & (5,3) & (6,3) \\
(1,4) & (2,4) & (3,4) & (4,4) & (5,4) & (6,4) \\
(1,5) & (2,5) & (3,5) & (4,5) & (5,5) & (6,5) \\
(1,6) & (2,6) & (3,6) & (4,6) & (5,6) & (6,6)
\end{array}
$$

- X can take values of $2, 3, \cdots, 12$.

x	2	3	4	5	6	7	8	9	10	11	12
$P(X = x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

- The probability mass function is $f(x) = P(X = x)$, and

$$f(x) = \begin{cases} 1/36, & \text{if } x = 2 \\ 2/36, & \text{if } x = 3 \\ \vdots \end{cases}.$$

- Clearly, $\sum_{i=1} f(x_i) = 1$.

2-2. Cumulative distribution function

- The **cumulative distribution function** (cdf) of a random variable X is defined as
$$F(x) = P(X \leq x), \quad \text{for } -\infty \leq x \leq \infty.$$

- This defintion applies to any random variables, discrete or continuous.

- Properties of the cdf of a random variable X

  1). Non-decreasing with x, i.e., if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$.

  2). $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$.

Finding probabilities from cumulative distribution functions

- For every value x, $P(X > x) = 1 - F(x)$.
- For all values x_1 and x_2 such that $x_1 < x_2$,
  $$P(x_1 < X \leq x_2) = F(x_2) - F(x_1).$$
- **Example**.

2-3. Independence of random variables

- Let X be a random variable, taking on values of $x_1, x_2, \cdots$; let Y be another random variable, taking on values of $y_1, y_2, \ldots$.
- X and Y are said to be **independent** if, for all i and j,
$$P(X = x_i \text{ and } Y = y_j) = P(X = x_i) \cdot P(Y = y_j).$$
- This definition applies to both discrete and continuous random variables.

2-4. Mean and variance of a discrete random variable

- Let X be a discrete random variable. The **expected value** or **mean** of X is
$$E[X] = \sum_{i=1}^{\infty} x_i P(X = x_i).$$

- We often use μ for $E[X]$. Sometimes we write EX instead of $E[X]$.

- If $Y = aX + b$ where a, b are finite constants, then
$$EY = aEX + b.$$

- For any two random variables X and Y,
$$E[X + Y] = EX + EY.$$

- The **variance** of any random variable X is the expected value of $(X - EX)^2$, that is, $\mathrm{var}(X) = E[(X - EX)^2]$.
- For a discrete random variable X,
$$\mathrm{var}(X) = \sum_{i=1}^{\infty} (x_i - \mu)^2 P(X = x_i).$$
- We often use σ^2 for $\mathrm{var}(X)$.

- It can be shown that
$$\mathrm{var}(X) = E[X^2] - (E[X])^2.$$
- If $Y = aX + b$ where a, b are finite constants, then
$$\mathrm{var}(Y) = a^2\mathrm{var}(X).$$
- If random variables X_1 and X_2 are independent, then
$$\mathrm{var}(X_1 + X_2) = \mathrm{var}(X_1) + \mathrm{var}(X_2).$$

3. Examples of discrete distributions

3-1. Geometric distribution

- In an experiment with two outcomes: success or failure, the probability of success is p. Repeat the experiment until a success occurs, the number of trials, N, is a discrete random variable. The distribution of N is
$$f(n) = P(N = n) = (1-p)^{n-1} \cdot p.$$
- Why? There are (n-1) trials with a failure, followed by a success in the n-th trial.
- Example. Tossing a coin. In each trial, the probability of a head is 1/2. The number of tosses until a first head follows a geometric distribution, $P(N = n) = \left(1 - \frac{1}{2}\right)^{n-1} \cdot \frac{1}{2}$.
- What is the probability of obtaining the first head in the 5th trial? $P(N = 5) = \left(1 - \frac{1}{2}\right)^4 \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^5 = 0.03125.$

3-2. Bernoulli distribution

- A random variable X that takes only two values 0 and 1 with $P(X = 1) = p$ has the **Bernoulli distribution with parameter $\boldsymbol{p}$**.
- We then can call X a Bernoulli random variable.
- The probability mass function of a Bernoulli random variable is $f(x) = (1-p)^{1-x}p^x$, with $x = 0$ or $x = 1$.
- In an experiment with two outcomes, we can code success to be 1 and failure to be 0.

3-3. Binomial distribution

- Suppose that n *independent* experiments (trials) are performed, and that each trial results in a "success" with probability p and a "failure" with probability $1-p$.
- The total number of successes, X, is a random variable. X has the **Binomial distribution with parameters n and p**.
- The probability of any specific sequence with k successes and $n-k$ failures is $p^k(1-p)^{n-k}$. There are $\binom{n}{k}$ such sequences, since there are $\binom{n}{k}$ ways to assign k successes to n trials.
- Therefore, the probability of k successes in n trials is obtained by **adding** the probability $p^k(1-p)^{n-k}$ for all $\binom{n}{k}$ sequences. That is,

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}.$$

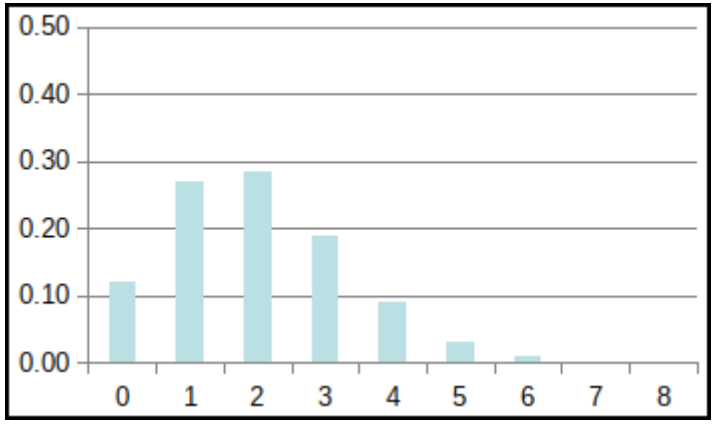
- $P(X = k)$ is indeed a distribution. First, $P(X = k) \geq 0$. Second, by the Binomial Theorem, $(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$, we have $\sum_{k=0}^{n} \binom{n}{k} p^k (1 - p)^{n-k} = 1$.

- **Example**. A machine produces a defective item with probability p. The number of defective items in n items follows a binomial distribution.

- **Example**. Tossing a coin 5 times, what is the probability of obtaining 2 heads? X is the number of heads. $p = 1/2$, $n = 5$, $k = 2$. Then $P(X = 2) = C_{5,2} \left(\frac{1}{2}\right)^2 \left(1 - \frac{1}{2}\right)^{5-2} = 0.3125.$

- The expected value of a binomial random variable X is
$$E[X] = \sum_{k=0}^{n} kP(X = k) = \sum_{k=0}^{n} kC_{n,k}p^k(1-p)^{n-k}.$$
- It turns out that $E[X] = np$.
- It can be shown that the variance of X is $\mathrm{var}(X) = np(1-p)$.

- Another example, $p = 0.1, n = 20$.



3-4. Hypergeometric distribution

- Suppose an urn contains n balls, of which r are black and $n-r$ are white. Let X be the number of black balls drawn when taking m balls without replacement.
- X then has the **hypergeometric distribution with parameters r, n, and m**,

$$P(X=k)=\frac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}}.$$

- Example?

3-5. Poisson distribution

- A discrete random variable X has the **Poisson distribution** with parameter $\lambda > 0$ if the probability mass function of X is
$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \cdots.$$
- here, $e = 2.71828$, sometimes called **Euler's number**. Fact: $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$.
- $\sum_{k=0}^{\infty} P(X = k) = 1$, because $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$.

- The Poisson distribution is the limit of the binomial distribution as the number of trials, n, approaches to infinity and the probability of success on each trial, p, approaches to zero in such a way that $np = \lambda$.

- Let $np = \lambda$, then $p = \frac{\lambda}{n}$, the binomial distribution function becomes

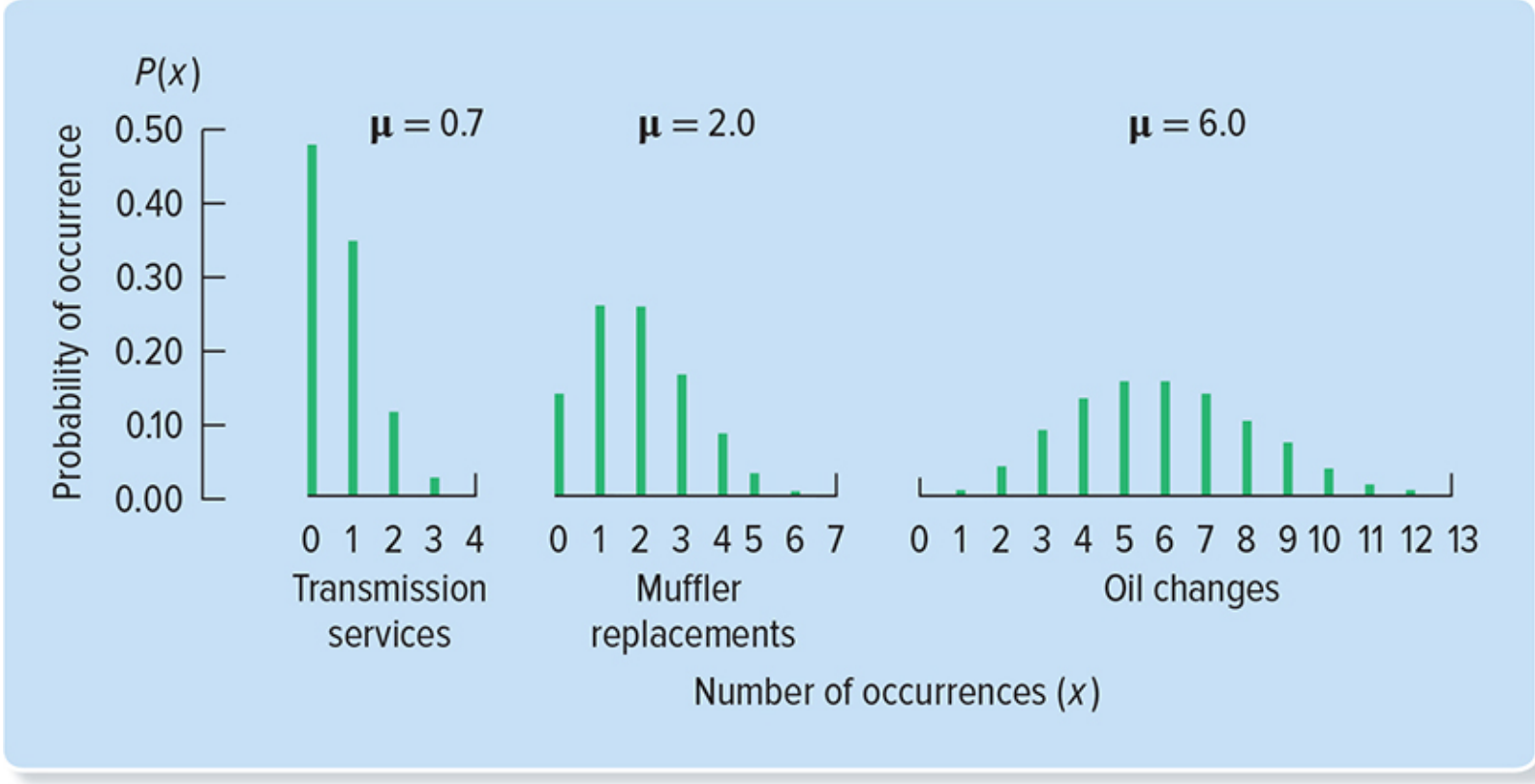
$$\begin{aligned} P(X = k) &= \frac{n!}{k!(n-k)!} \cdot \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \cdot \frac{n!}{(n-k)!n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k}. \end{aligned}$$

- As $n \to \infty$, $\left(1 - \frac{\lambda}{n}\right)^n \to e^{-\lambda}$, $\frac{n!}{(n-k)!n^k} \to 1$, and $\left(1 - \frac{\lambda}{n}\right)^{-k} \to 1$.

- Therefore, as $n \to \infty$, $P(X = k) \to \frac{\lambda^k e^{-\lambda}}{k!}$ which is Poisson.

- The expected value of a Poisson random variable X is
$$E[X] = \lambda.$$
- Recall that the mean of a binomial variable is np. And for a Poisson random variable X, $\lambda = np$.
- Parameter λ can be considered as the average number of occurrences.
- **The variance of $\boldsymbol{X}$ is also $\boldsymbol{\lambda}$**. Skipping the derivation.

- Some examples.



- Thus, we can use the Poisson distribution to approximate the binomial distribution for large n and small p.
- Poisson distribution can be used in the following situation.
- Let X be a random variable that equals the number of occurrences of some event in a given time interval. We can think of dividing the time interval into a large number of small sub-intervals of equal length, and assume that the sub-intervals are so small that the probability of more than one event in a sub-interval is negligible relative to the probability of one event which is itself very small.
- Also assume that the probability of an event is the same in each sub-interval, and whether an event occurs in one sub-interval is independent of occurring in other sub-intervals.
- X then has approximately a Poisson distribution.

- Poisson distribution is important in practice.
- It is widely used in modeling the **count data**.
- Examples
    - The number of emails you receive in a day.
    - The number of customers visiting a bank branch in an hour.
    - The number of patents a received annually by a firm.
    - The number of insurance claims received in an insurance company.

4. Continuous random variables and their distributions

- A random variable is **continuous** if it takes on a continuum of values rather than a finite number of values.
- A random variable X is said to have a continuous distribution with **probability density function** or simply **density function** f if for all $a \leq b$, we have
$$P(a \leq X \leq b) = \int_a^b f(x)dx.$$
- Draw a graph showing $P(a \leq x \leq b)$.
- A density function $f(x)$ satisfies: i) $f(x) \geq 0$; ii) $\int_{-\infty}^{\infty} f(x) = 1$.

- Since X takes on a continuum of values, the probability of a single value x is zero, that is, $P(X = x) = 0$. $P(X = x) \neq f(x)$.

- To see this, let $a = x$ and $b = x + \Delta x$ with $\Delta x > 0$ but small. Then
$$P(x \leq X \leq x + \Delta x) = \int_x^{x+\Delta x} f(z)dz \approx f(x)\Delta x.$$
As $\Delta x \to 0$, $P(X = x) \to 0$.

- But $f(x)$ tells the likelihood of X being near x. That is $\frac{P(x \leq X \leq x+\Delta x)}{\Delta x} \approx f(x)$.

- **Cumulative distribution function** of a continuous random variable X is
$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(z)dz.$$
- The probability that X falls into an interval $[a, b]$ is
$$P(a \leq X \leq b) = \int_{a}^{b} f(x)dx = F(b) - F(a).$$

- The **expectation** of a continuous random variable X is
$$E[X] = \int_{-\infty}^{\infty} x f(x) dx.$$
- The expectation is also called the **mean** or **expected value**. We also use $\mu = E[x]$.
- The variance of X is
$$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx.$$

4-1. Uniform distribution

- A **uniform random variable** X on the interval $[a, b]$ has the density function
$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{for } a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$
- The cumulative distribution function of X is $P(X \leq x) = \frac{x-a}{b-a}$ if $x \in [a, b]$.
- The mean of a uniform random variable is $\mu = \frac{a+b}{2}$.
- The variance of a uniform random variable is $\sigma^2 = \frac{(b-a)^2}{12}$.

4-2. Normal distribution

- A **normal random variable** X has the following probability density function
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$
- Its mean is μ and its variance is σ^2.